ECON 164: Theory of Economic Growth

Week 9A: Natural Resources and the Environment

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Introduction

- **last time**: added factor in **fixed supply** (**land X**) $\longrightarrow$ declining GDPpc unless...
  - ...endogenous technological change A wins the race against declining X/L
  - but we treated land as a **renewable resource**: can use the same land over and over

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  - ...endogenous technological change A wins the race against declining X/L
  - but we treated land as a **renewable resource**: can use the same land over and over
- **today**: **nonrenewable resources** (petroleum, coal, natural gas, minerals, ...)
  - like last time: race between A vs. X/L *and* E/L
  - quantify the **drag on growth** from **resource extraction**
  - consider relationship between **GDP per capita** and **pollution**

Production with nonrenewables

Production now also requires **land** $\boldsymbol{X}$ and **energy** $\boldsymbol{E}$:

$$Y_t = K_t^{\alpha} \boldsymbol{E_t}^{\beta} \boldsymbol{X}^{\gamma} (A_t L_t)^{1-\alpha-\beta-\gamma}$$

notice: **constant returns to scale** in $\{K, \boldsymbol{E}, \boldsymbol{X}, L\}$

Capital, labor, and technology growth as in the **Solow** model:

$$\dot{K}_t = s_I Y_t - \delta K_t$$
$$\dot{L}_t = g_L L_t$$
$$\dot{A}_t = g_A A_t$$

Resource extraction

To get **energy** E, need to extract **nonrenewable resource** w/ initial stock R_0:

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How much to extract? For now, just set fraction s_E, so... (Dasgupta and Heal, 1974)

$$\frac{\dot{R}_t}{R_t} = -\frac{E_t}{R_t}$$

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$$g_R = -s_E \quad \rightarrow \quad g_E = -s_E$$

because $E_t = s_E R_t$

How do we solve this model?

Recall the production function:

$$Y_t = K_t^{\alpha} \boldsymbol{E_t}^{\beta} \boldsymbol{X}^{\gamma} (A_t L_t)^{1-\alpha-\beta-\gamma}$$

If we could instead write...

$$Y_t = K_t^{\alpha} (\boldsymbol{B_t} L_t)^{1-\alpha}$$

...**would we know what to do?**

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...**would we know what to do?**

Yes! This is just the **Solow model** with **labor-augmenting productivity $\boldsymbol{B_t}$**...

- steady state where $g_{\tilde{k}} \equiv g_K - \boldsymbol{g_B} - g_L = 0 \rightarrow$ **phase diagram**

$$\tilde{k}^{\text{ss}} = \left(\frac{s_I}{\boldsymbol{g_B} + g_L + \delta} \right)^{\frac{1}{1-\alpha}}$$

- slope of BGP: $g_y^{\text{ss}} = \boldsymbol{g_B}$

What is B_t?

Taking ratios of the two possible forms of the production function...

$$1 = \frac{K_t^{\alpha} \boldsymbol{E_t}^{\beta} \boldsymbol{X}^{\gamma} (A_t L_t)^{1-\alpha-\beta-\gamma}}{K_t^{\alpha} (\boldsymbol{B_t} L_t)^{1-\alpha}} = \frac{\boldsymbol{E_t}^{\beta} \boldsymbol{X}^{\gamma} A_t^{1-\alpha-\beta-\gamma} L_t^{-(\beta+\gamma)}}{\boldsymbol{B_t}^{1-\alpha}}$$

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...and its growth rate is then

$$\boldsymbol{g_B} = \frac{1-\alpha-\beta-\gamma}{1-\alpha} g_A + \frac{\beta}{1-\alpha}(\boldsymbol{g_E} - g_L) + \frac{\gamma}{1-\alpha}(\boldsymbol{g_X} - g_L)$$

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Defining the resource drag on growth

We can rearrange the expression for the growth rate of $\boldsymbol{B_t}$ to get...

$$g_y^{\text{ss}} = \boldsymbol{g_B} = \left(1 - \frac{\beta+\gamma}{1-\alpha}\right) g_A - \underbrace{\frac{\beta}{1-\alpha}(\boldsymbol{s_E} + g_L)}_{\text{drag from energy } \boldsymbol{E}} - \underbrace{\frac{\gamma}{1-\alpha} g_L}_{\text{drag from land } \boldsymbol{X}}$$

- both include spreading finite resources over more people...
- ...but the drag from **energy** $\boldsymbol{E}$ also includes the depletion of the resource

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Recall: We saw this kind of resource drag at the end of the Malthusian model, too

$$g_y^{\text{ss}} = \beta(g_A^{\text{ss}} - g_L^{\text{ss}}) = (1-\beta)\frac{\lambda}{1-\phi} g_L^{\text{ss}} - \underbrace{\boldsymbol{\beta g_L^{\text{ss}}}}_{\text{drag from land } \boldsymbol{X}}$$

Quantifying the resource drag on growth (Nordhaus, 1992)

- $\alpha \equiv$ share of total factor payments to **capital** K ≈ 0.2
- $\beta \equiv$ share of total factor payments to **energy** E ≈ 0.1
- $\gamma \equiv$ share of total factor payments to **land** X ≈ 0.1

$\rightarrow$ the **labor share** is then ≈ 0.6 like the Kaldor facts!

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$$-\frac{\beta}{1-\alpha}(s_E + g_L) - \frac{\gamma}{1-\alpha}g_L = -\left(\frac{0.1}{1-0.2}\right)(\mathbf{0.005} + 0.01) - \left(\frac{0.1}{1-0.2}\right)0.01$$
$$= (1/8) \times 0.015 + (1/8) \times 0.01 \approx \mathbf{0.003}$$

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$\rightarrow$ **1.8% observed growth** includes **0.3% drag** from **energy** and **land**

Comparative statics with respect to s_E

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Say we permanently decrease s_E at time T $\longrightarrow$ just like a permanent **increase** in g_B

1. What happens to $\tilde{k}^{\text{ss}}$? (**draw the phase diagram!**)
2. What happens to g_y^{ss} at T? In the long run?
3. What happens to $\ln y_t^{\text{BGP}}$ at T? In the long run?

(check: **Problem Set #2, Question 2b**)

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- **what's the optimal s_E?** think about analogy w/ s_I and s_R:
  - s_I in **NGM**: more consumption today vs. more consumption tomorrow
  - s_R in **(semi-)endogenous**: more output today vs. more/better ideas for tomorrow

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- **but something's different**: s_E, unlike s_I and s_R, *does* affect the growth rate g_y^{ss}
- **why?** depletion of R is a linear differential equation $\longrightarrow$ like **AK models**

How to value "energy today" vs. "resources for tomorrow"

Suppose the representative household has preferences

$$U_t = (c_t - \bar{c})^{1-\rho} R_t^{\rho}$$

and faces the *implicit* budget constraint

$$y_t = c_t + p_{Rt} R_t$$

- $\bar{c}$ is **subsistence consumption**, like in the Malthusian model
- household cares directly about the **stock of resources** R_t
  - if **resource extraction** implies **pollution** $\rightarrow$ captures the value of clean environment
  - can reinterpret E_t as **emissions**
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Solving the household's problem

Compute the marginal utility of consumption...

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Production and prices

To simplify, let's set $\alpha = \gamma = 0$ to eliminate **capital** (no saving) and **land**...

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Note the tradeoff: more **energy** $\rightarrow$ more **output** $\boldsymbol{y_t}$ but lower **resources** $\boldsymbol{R_t}$

$$p_{Rt} = \frac{\partial y_t}{\partial E_t}$$

The optimal choice of s_E

Take the household's optimality condition and the price of resources...

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Take the household's optimality condition and the price of resources...

$$\frac{\rho}{1-\rho}\left(\frac{c_t-\bar{c}}{R_t}\right)=\beta\frac{y_t}{E_t}$$

...then rearrange to get the optimal choice of s_E:

$$s_{Et}\equiv\frac{E_t}{R_t}=\beta\left(\frac{1-\rho}{\rho}\right)\frac{y_t}{y_t-\bar{c}}$$

Economic growth leads to better environmental quality

$$s_{Et} = \beta \left(\frac{1-\rho}{\rho} \right) \frac{y_t}{y_t - \bar{c}}$$

- when y_t is close to subsistence level $\bar{c}$...
  - $y_t/(y_t - \bar{c})$ is high
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  - **think**: electric cars, recycling, low-impact products, ...

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- better model would predict "environmental Kuznets curve" (Brock and Taylor, 2010)
  - s_{Et} rises with y_t at low levels, then falls at high levels

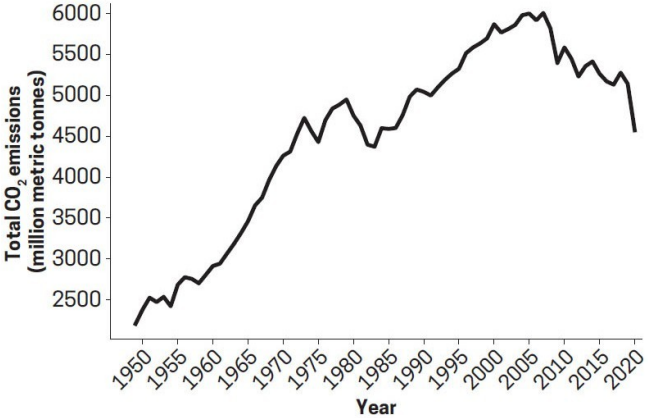
Does economic growth lead to lower resource extraction?

Figure 10.5 Energy Use per Capita in the United States, 1950–2019

SOURCE: Authors' calculations from U.S. Energy Information Administration (2021*a*).

Does economic growth lead to lower pollution?

Figure 10.6 Total CO_2 Emissions in the United States, 1950–2019



SOURCE: Authors' calculations from U.S. Energy Information Administration (2021a).

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Figure 10.7 Renewables as a Percent of Energy in the United States, 1950–2019

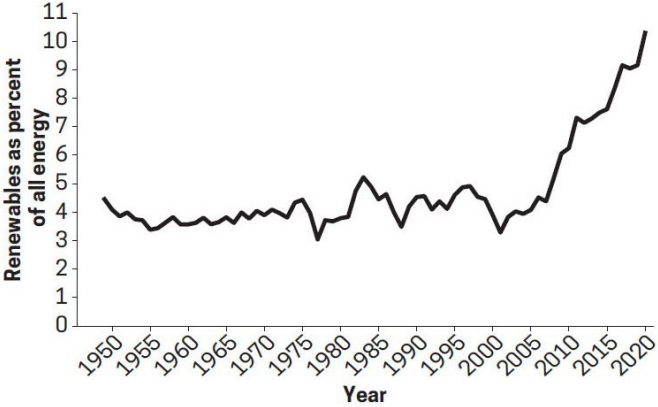

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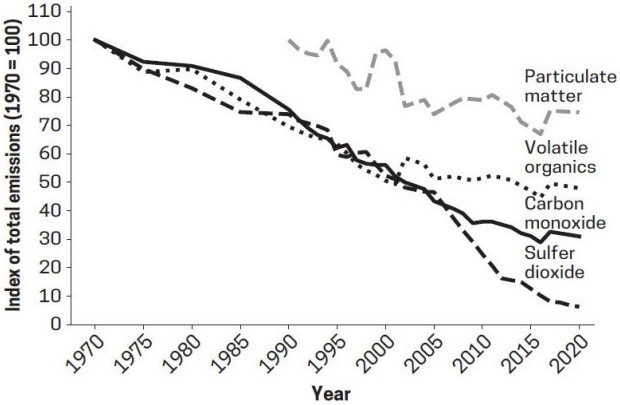
Figure 10.8 Air Pollutants in the United States, 1970–2019



SOURCE: Authors' calculations from U.S. Environmental Protection Agency (2021).

NOTE: The indices for particulate matter are indexed to 100 in 1990, given a lack of prior data. All other series are indexed to 100 in 1970.

Summary

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  - ...endogenous technological change A wins the race against declining X/L
  - but we treated land as a **renewable resource**: can use the same land over and over
- **today**: **nonrenewable resources** (petroleum, coal, natural gas, minerals, ...)
  - like last time: race between A vs. X/L *and* E/L
  - can analyze like **Solow model** with **labor-augmenting productivity** B_t
  - find **0.3% drag on growth** from **land** and **resource extraction**
  - some evidence that higher **GDP per capita** leads to lower **emissions** s_E

References

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